

DAI1000

Digital and AI Literacy

L4 - AI and Technology Convergence

Class Objectives

- Three Intellectual Traditions
 - Symbolic AI
 - Statistical AI
 - Neural AI
- AI and other Technologies

Three Intellectual Traditions



Symbolic AI

Logic-based approach

Using symbols and rules to represent knowledge and reasoning



Statistical AI

Data-driven approach

Using statistical methods and probabilistic models to make predictions



Neural AI

Brain-inspired approach

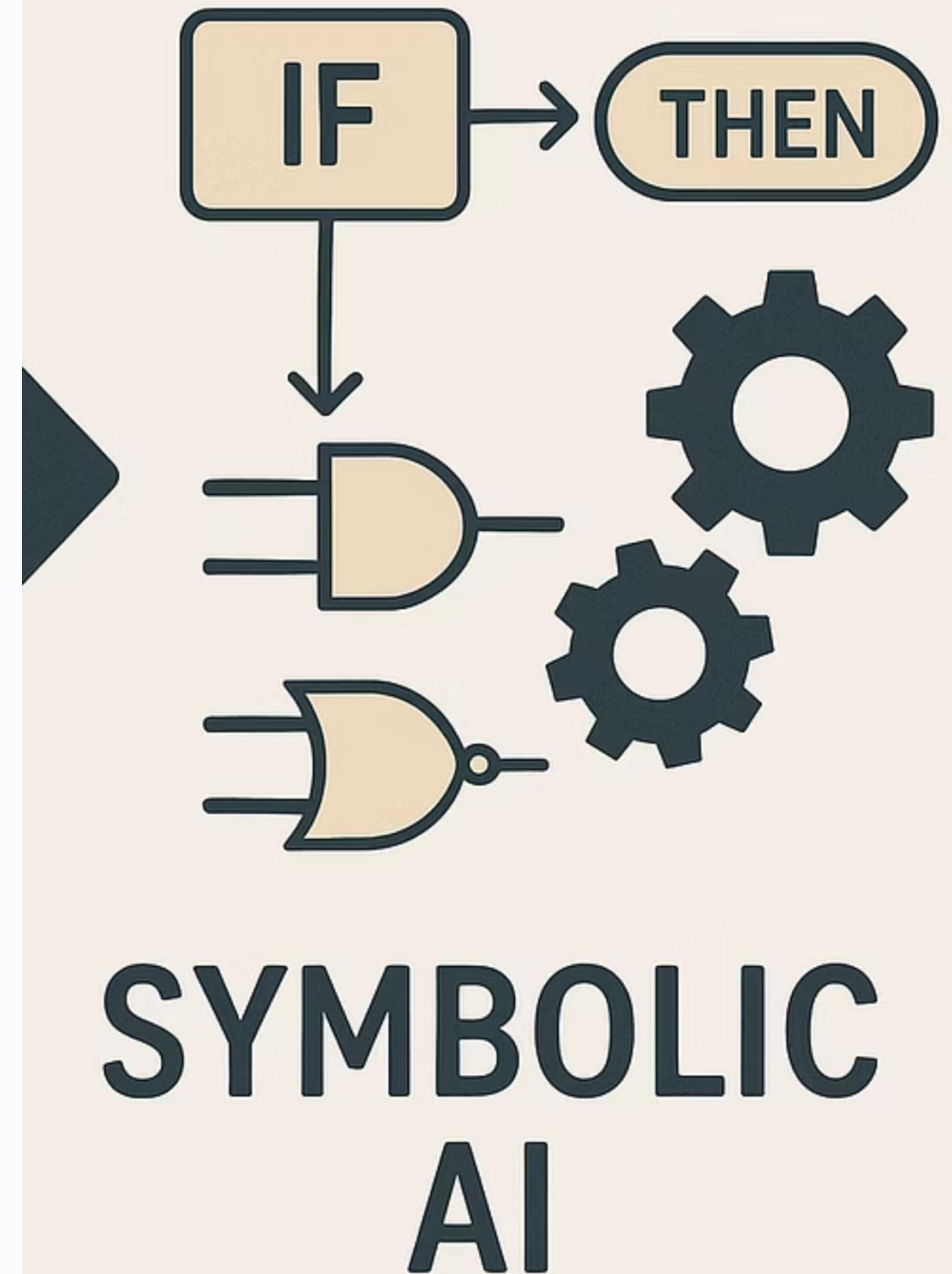
Using artificial neural networks to learn patterns and make decisions

Symbolic AI

Pioneered by John McCarthy and Marvin Minsky in the 1950s and 1960s

Human knowledge can be explicitly represented using symbols and logical rules, much like mathematics or formal logic

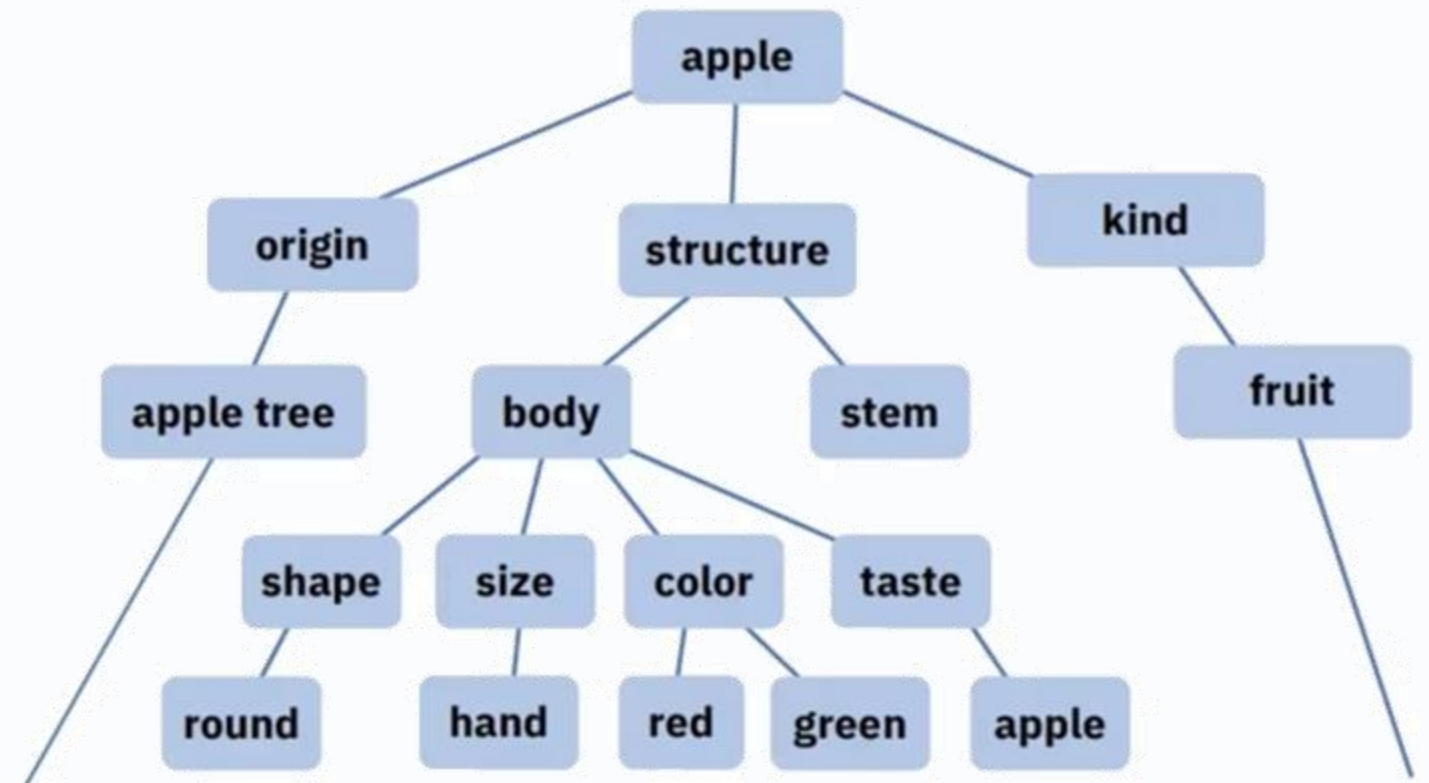
Machines could reason and solve complex problems by manipulating symbols and rules systematically



Symbolic AI

- ☐ A: cloudy
- B: humidity > 70%
- T: about to rain

A & B -> T



Example: Logic Theorist (1956)

Prove mathematical theorems using symbolic logic and heuristic reasoning

01

Axioms Foundation

Started with a set of axioms
- known mathematical truths that serve as the foundation

03

Step-by-Step Reasoning

Figured out how to reach target statements (theorems) through logical progression

02

Rules of Inference

Applied logical rules to derive new truths from existing ones systematically



Example: ELIZA (1966)—The Conversational Pioneer

ELIZA demonstrated symbolic AI's potential through surprisingly human-like conversations using pattern matching and rule-based responses.

Input Processing

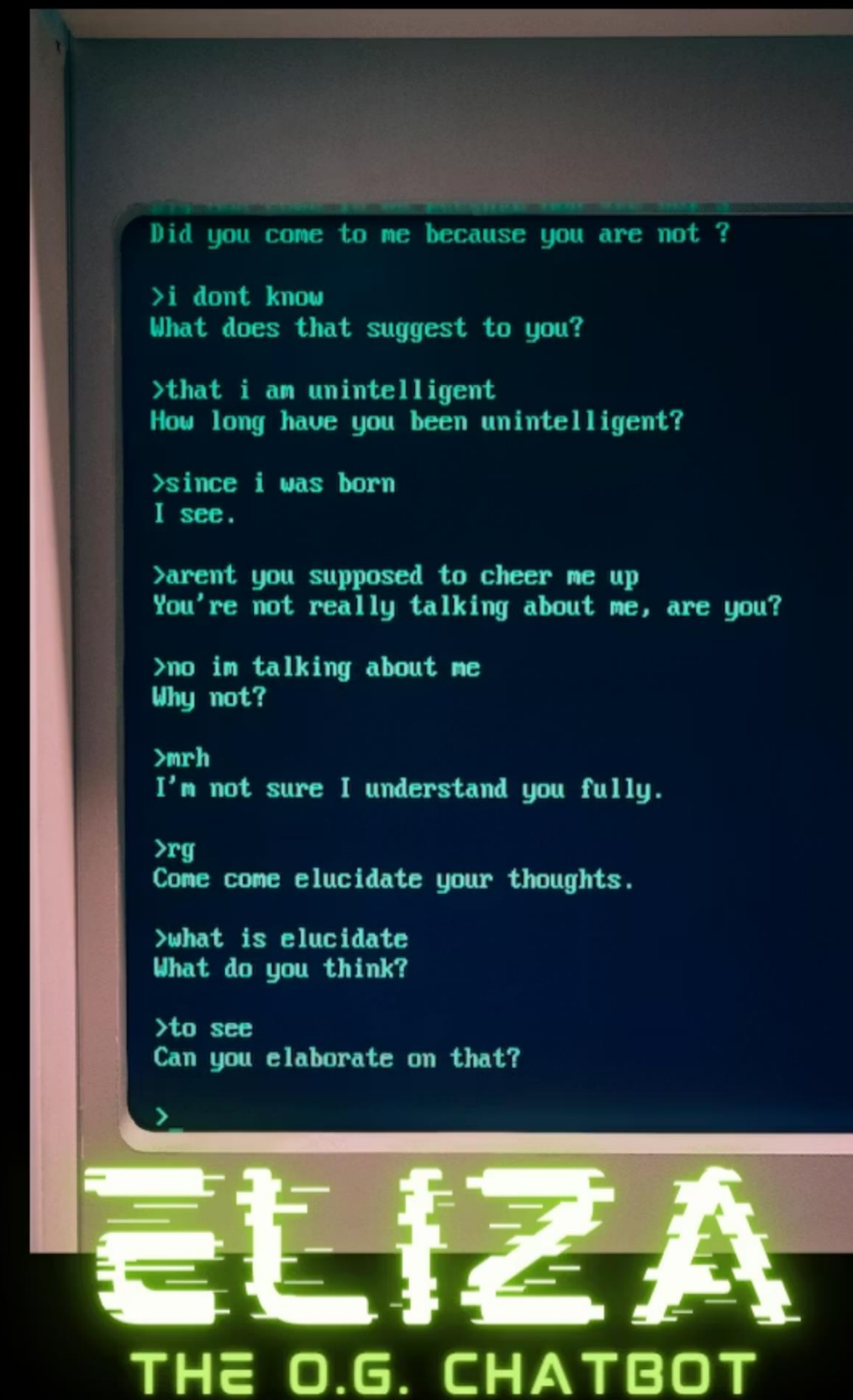
Takes user input and searches for specific keywords or phrases

Pattern Matching

Applies simple rules based on syntactic structures

Response Generation

The output is a reformulation of the input, often transformed into a question. If no keywords are detected, ELIZA resorts to generic responses such as “Tell me more” or “Why do you say that?”



Example: Expert Systems

Expert systems represent another branch of symbolic AI, using handcrafted "if-then" rules provided by human experts to make inferences.

Symbol Representation

Use symbols to represent facts and logical relationships between concepts

Human Expert Knowledge

Rely on rules crafted by domain experts to replicate human decision-making

Medical Example

IF patient has fever **AND** cough **THEN** diagnosis is flu

Symbolic AI: Strengths & Limitations

Strengths

Transparent and explainable: Each decision could be traced back to a specific rule, making them trustworthy and auditable.

Limitations

- Became less precise when more rules were added
- Costly and needed constant updating
- Inflexible and failed to adapt to new situations

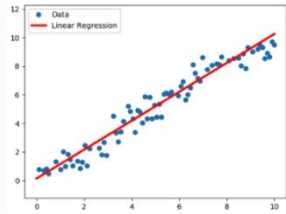
Statistical AI: The Data-Driven Revolution

As computing power increased and datasets became more available, AI research shifted towards statistical methods in the 1980s.

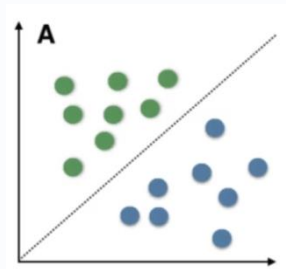
It allowed computers to **learn from data, recognizing patterns, and make predictions based on statistical probabilities.**

Statistical AI methods include linear regression, support vector machines, decision trees, bayesian networks and so on

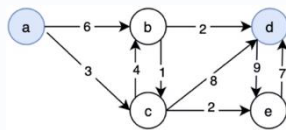
Statistical AI



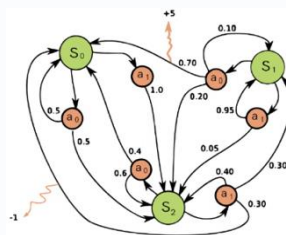
- 1801: linear regression (Gauss, Legendre)



- 1936: linear classification (Fisher)



- 1956: Uniform cost search for shortest paths (Dijkstra)



- 1957: Markov decision processes (Bellman)

Statistical AI Example: Linear Regression

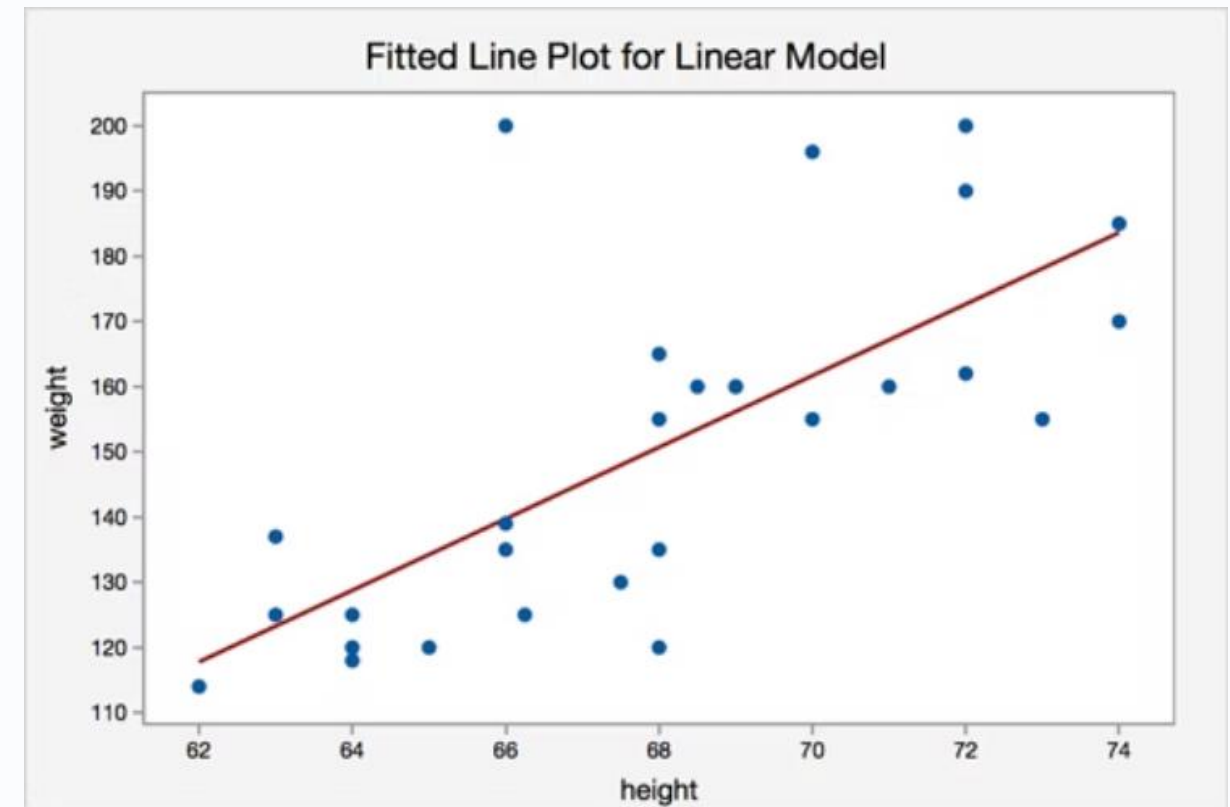
Enable machines can learn to predict continuous outcomes by identifying relationships within data.

Learning a linear relationship between an input variable (e.g., height) and an output variable (e.g., weight), finding the "best-fit" straight line

For instance, a simple linear model might predict a person's weight based on their height using a formula like:

$$\text{Weight} = 0.9 \times \text{Height} - 85$$

Prediction: If a person is 175 cm tall, their predicted weight would be: $0.9 * 175 - 85 = 157.5 - 85 = 72.5$ kg.



Statistical AI: Trade-offs

Strengths

Generalization & Adaptability

Better generalization from examples, not limited to specific domains

Less Domain Knowledge

Required less domain-specific expertise compared to symbolic AI

Continuous Improvement

Gradually improve performance through experience

Limitations

Requires human effort

Requires human experts to decide what features to use.

Limited to Simple Patterns

Is great for structured data (e.g., tabular data), but they struggle with raw, unstructured data like: images audio, and natural language

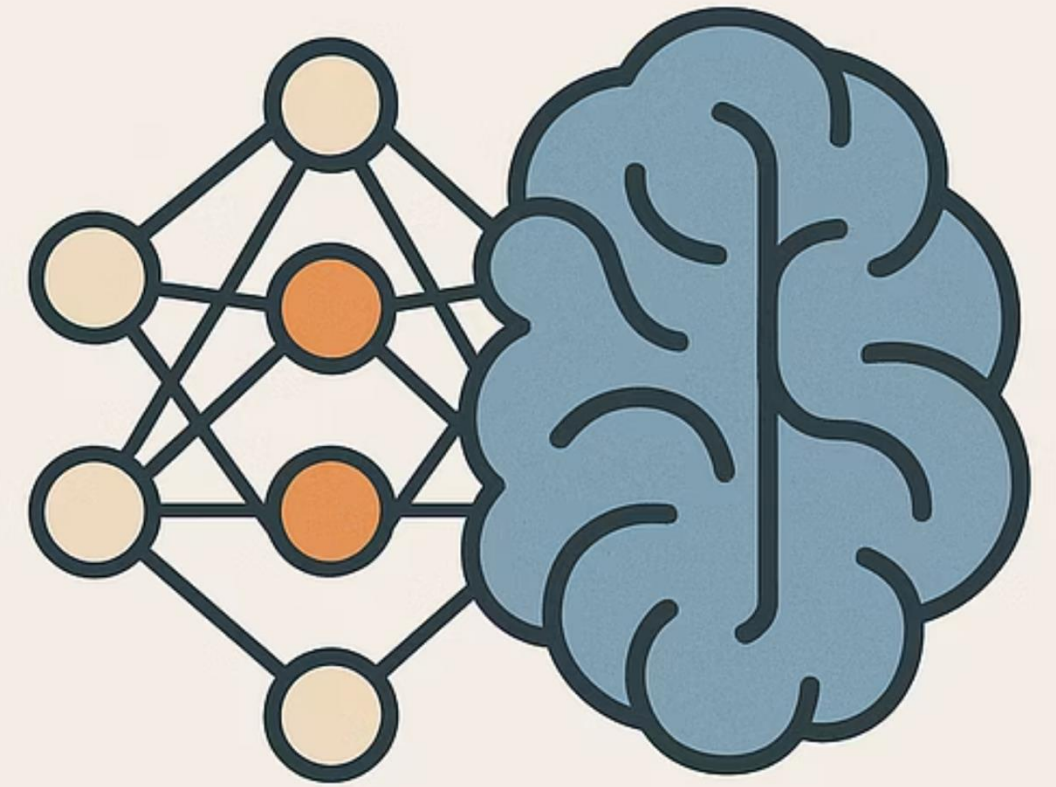
Scaling Issues with Complexity

It becomes less effective when the data has many interacting variables and the problem has nonlinear patterns

Neural AI

From the end of the last century to present, Neural AI developed from Statistical AI foundations, utilizing Artificial Neural Networks for learning and reasoning.

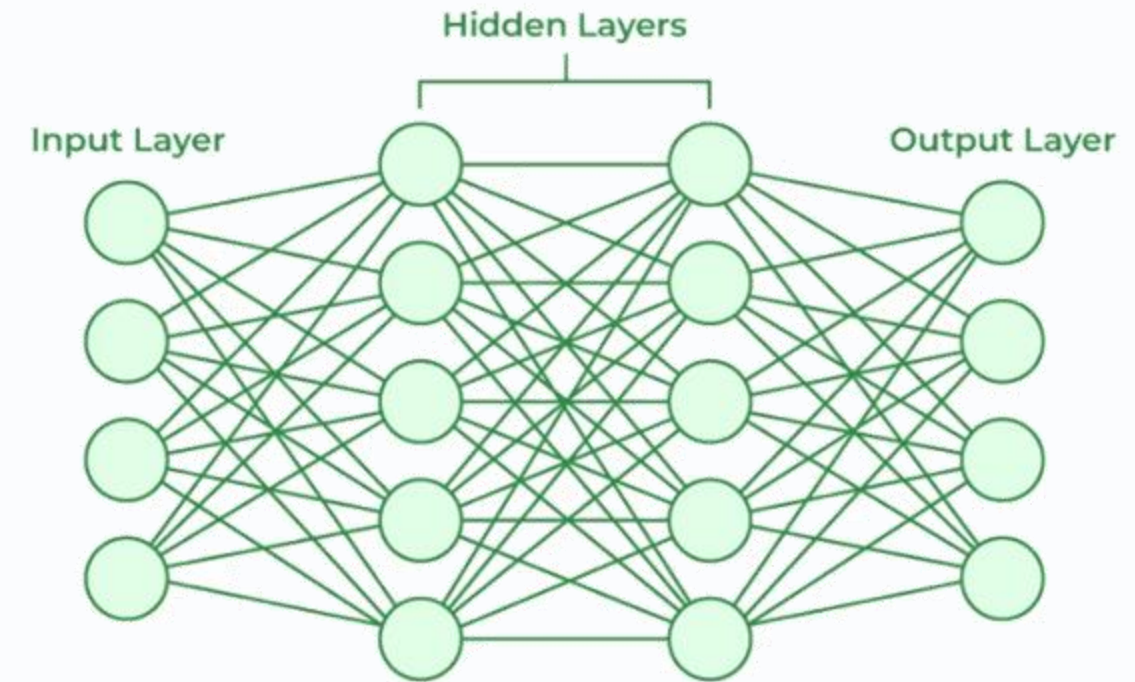
Is inspired by the structure and function of the human brain.



NEURAL NETWORKS

Neural AI

- Taking input data (e.g., an image, sound, or sentence)
- Passing it through layers of neurons, where each neuron makes a small calculation
- Adjusting the strength of connections to learn patterns
- Producing an output (e.g., “this is positive sentiment”)



Neural AI: Power & Pitfalls

Strengths

Address Complex Issue

Capture nonlinear, hierarchical, and abstract patterns, making it suitable for complex tasks like image recognition or natural language understanding

Feature Engineering

Manually designing or selecting the right input features for a model to learn effectively

Limitations

Still Brittle

Fail to generalize beyond training data, vulnerable to adversarial examples

Black Box

Challenging to understand how they arrive at specific decisions

Three Intellectual Traditions

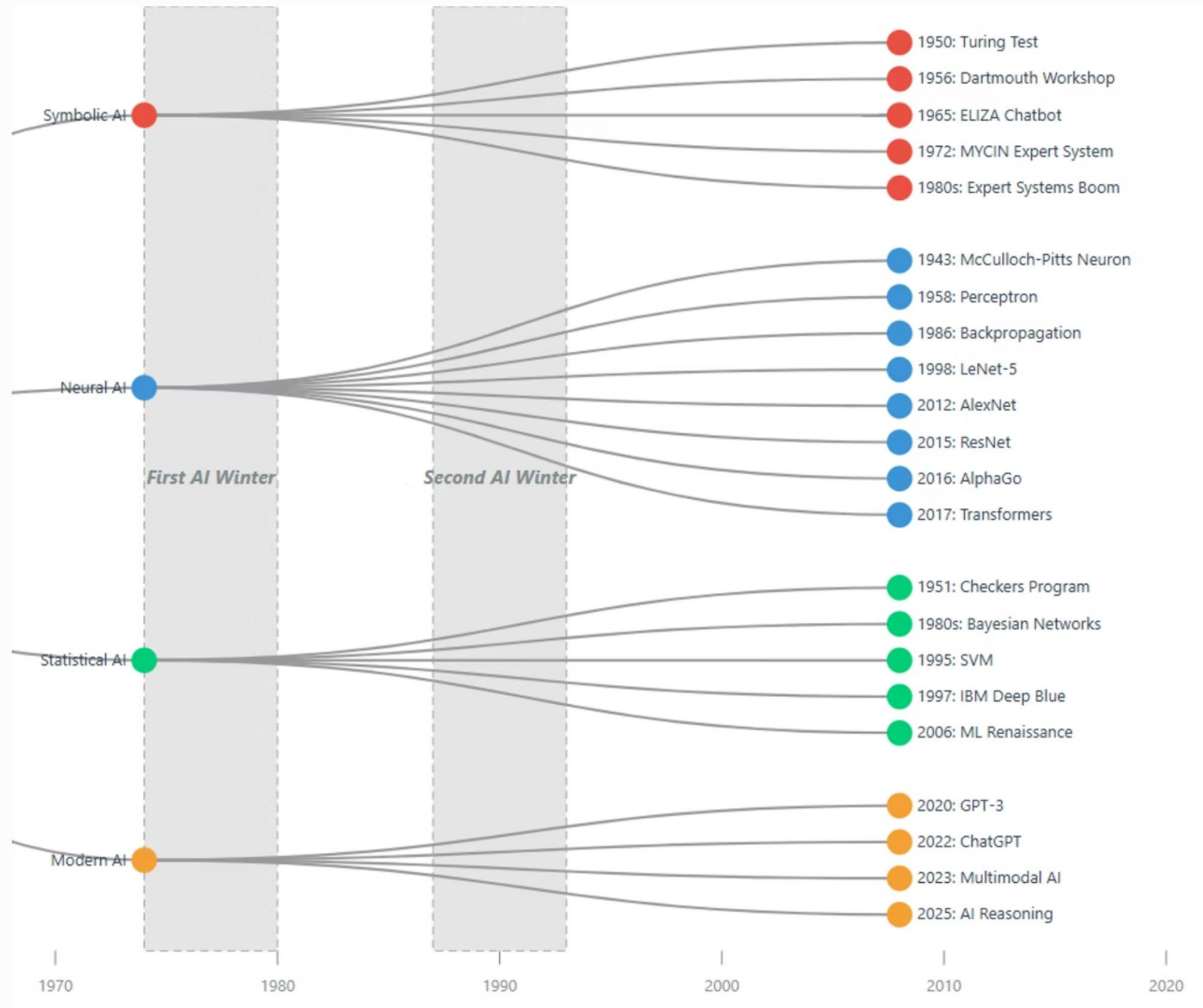
- **Symbolic AI** took a **top-down** approach and failed to fulfill its original promise. But it offered a vision and did build impressive artifacts for ambitious problems like question answering and dialogue systems
- **Neural AI** took a different approach (**bottom-up**), starting with simple perceptual tasks

Three Intellectual Traditions

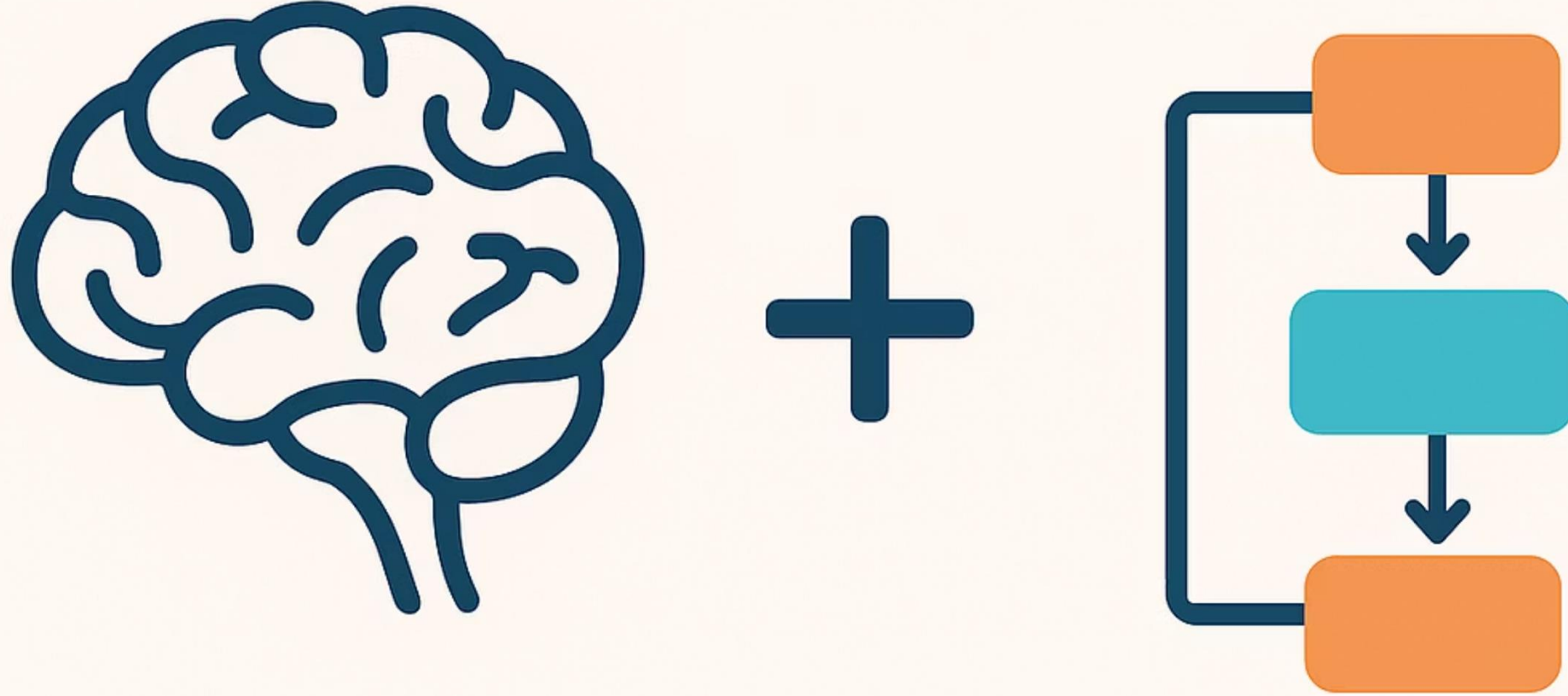
- **Statistical AI** foremost offers mathematical rigor and clarity
 - For example, we define an objective function separate from the optimization algorithm, or have a language to talk about model complexity in learning

The AI world is like a melting pot that has drawn from many different fields ranging from statistics, algorithms, neuroscience, optimization, economics, etc.

It is the symbiosis between these fields and their application to important real-world problems that makes working in the field of AI so rewarding.



NEURO-SYMBOLIC AI



Future Direction: Neuro-Symbolic AI

Integrating neural networks with symbolic reasoning creates hybrid systems that combine robust learning with interpretability and logical structure.

- Neural Enhancement

Neural nets split the world into symbols, recognizing images and translating pixels into symbolic representations

- Symbolic Injection

Symbolic algorithms inject common sense reasoning and domain knowledge into neural networks. They would apply logic and semantic reasoning to describe relationships between objects, predict various interactions, answer questions and make decisions

Class Objectives

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 - Statistical AI
 - Neural AI
- AI and other Technologies

Technologies that Impact AI

- Many technologies that influence and improve AI capabilities are reviewed
- Technologies and new capabilities have increasingly converged over time (e.g., robots and autonomous vehicles)

Fully Developed Technologies

Cloud Computing

- Dominant in global IT infrastructure (AWS, Azure).
- AI services (e.g., SageMaker, Vertex AI) are commoditized.
- Scalable and cost-efficient for AI workloads.

GPS/Satellites

- Decades of refinement; integral to logistics, navigation, and defense.
- AI-enhanced geospatial analytics (e.g., Google Maps) are mainstream.
- Low barriers to integration.

Computer Vision

- Mature frameworks (OpenCV, TensorFlow).
- Mass adoption in security, retail, and healthcare.
- High accuracy in controlled environments.

NLP

- Transformers (e.g., BERT, GPT) revolutionized language tasks.
- Ubiquitous in chatbots, translation, and sentiment analysis.
- Commercial APIs readily available.

3D Modeling

- Standard in gaming, architecture, and manufacturing.
- AI automates rendering, simulation, and design optimization.
- Tools like Blender have democratized access.

Still-Evolving Technologies

Edge Computing

- Growing but not yet ubiquitous due to hardware constraints.
- AI use cases (e.g., autonomous drones) are emerging.
- Standards (e.g., 5G MEC) are evolving.

Wearables

- Rapid innovation (e.g., smartwatches, AR glasses).
- AI integration is experimental (e.g., real-time health diagnostics).
- Battery life and data privacy are unresolved.

Robotics

- Industrial robots are mature, but AI-driven robotics (e.g., adaptive manufacturing) is nascent.
- High costs limit scalability.
- Ethics of human-robot collaboration are debated.

Digital Twins

- Conceptually powerful but implementation is complex.
- AI-driven simulations are pilot-stage in industries like energy and aerospace.
- Requires massive data integration.

Data-in-Motion

- Real-time streaming (e.g., Kafka, Flink) is advancing.
- AI for instant analytics is promising but resource-intensive.
- Latency and security challenges persist.

Emerging Technologies

Quantum Computing

- Labs-only stage (IBM, Google).
- Theoretical AI applications lack practical scalability.
- Extreme technical barriers (cryogenic cooling, error rates).

Neuromorphic Computing

- Mimics brain architecture; prototypes exist (Intel Loihi).
- Potential for ultra-efficient AI but no commercial deployment.
- R&D-focused with high costs.

Advanced Robotics

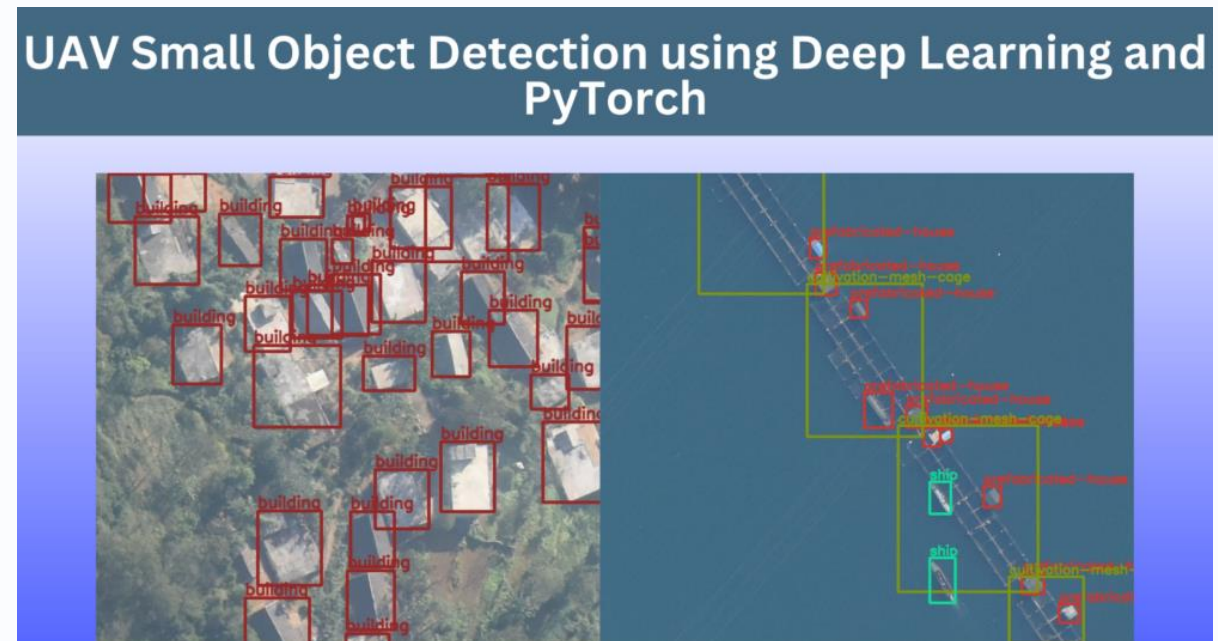
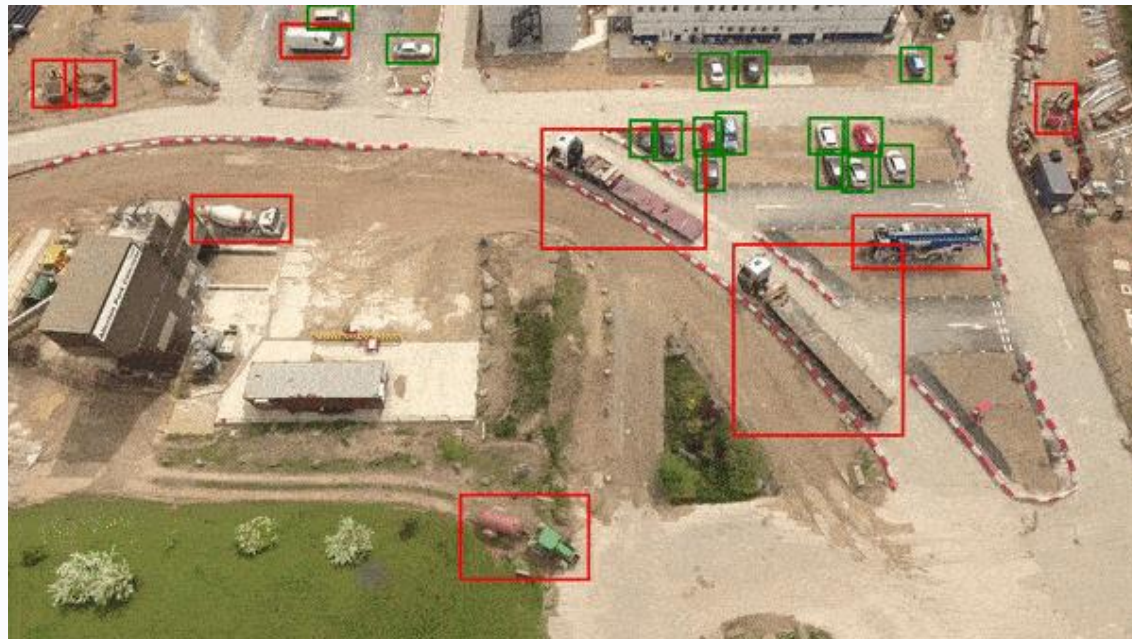
- Humanoid robots (e.g., Boston Dynamics) are demos, not products.
- AI collaboration (e.g., surgical robots) is in trials.
- Ethical and safety concerns dominate.

AI and GPS/Satellites

- AI's role in climate research has been transformative. Vast and intricate geospatial data are obtained from satellites, aerial photography, and GPS for AI analysis.
- **Predictive analysis:** AI-powered geospatial analytics play a vital role in predicting the path of typhoon or the spread of wildfires
- **Agricultural adaptation:** Climate-smart agriculture relies on AI to predict weather patterns, analyze soil health, and suggest optimal planting/harvesting efforts
- **Urban planning:** Design flood defenses and plan urban landscapes to mitigate the impacts of heavy rainfall and rising sea levels, land-use changes, deforestation ...

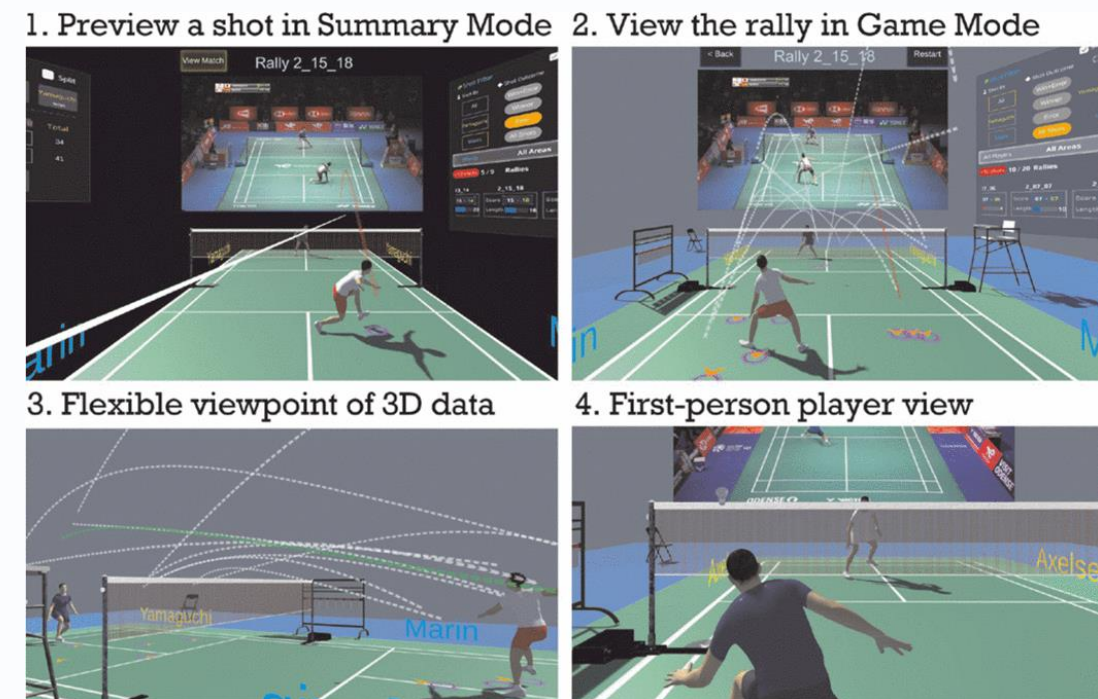
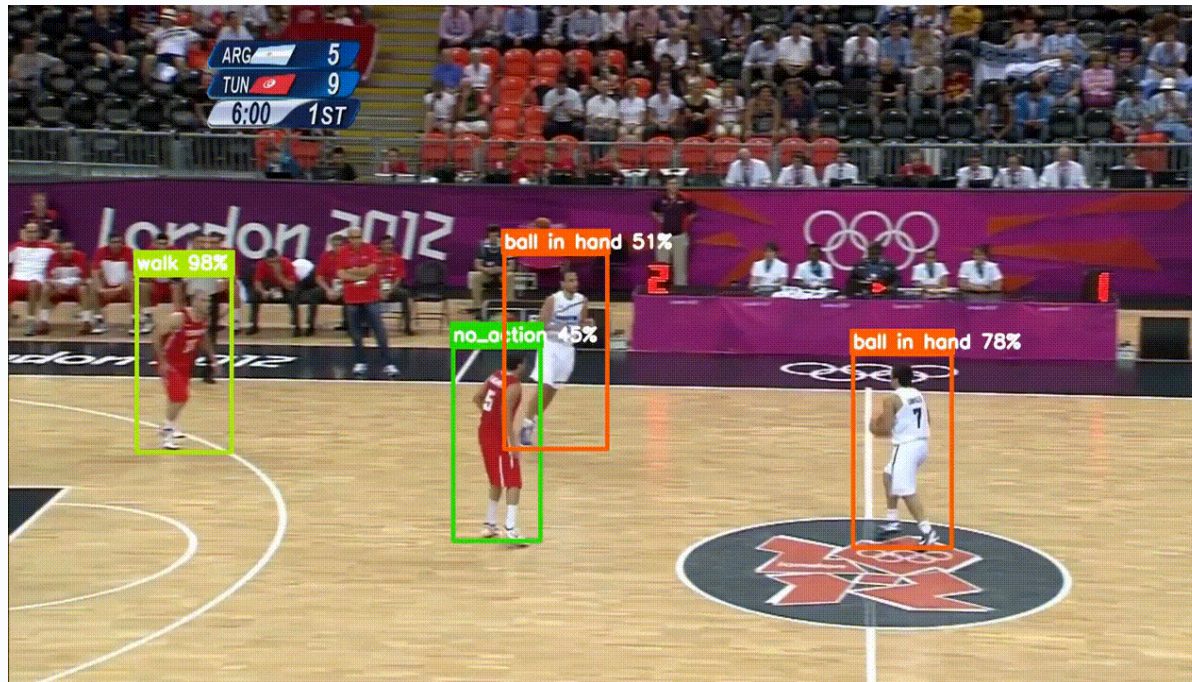
AI and Drones

- The integration of AI into drones offers numerous advantages
 - **Autonomy**: AI algorithms enable drones to operate **autonomously**
 - **Data processing**: Drones equipped with AI can process and analyze data in real-time
 - **Adaptive behavior**: AI allows drones to **adapt** to changing conditions
 - **Increased safety**: AI helps drones detect and avoid obstacles, reducing collision risks



AI and Computer Vision

- **Computer Vision**: perceive and comprehend images and videos, surpassing the capabilities of human eyes
- Computer Vision is revolutionizing **sports**: track and analyze the movements of players to offer invaluable **data on speed and technique**. Interactive elements enhance the delivery of player and ball tracking data, enabling analysis that offers fans **immersive experiences**



AI and 3D Modeling

- The integration of Artificial Intelligence (AI) into 3D model-maker tools has revolutionized the design landscape. AI algorithms enhance efficiency by automating repetitive tasks and streamlining workflows
- Machine learning algorithms, for example, can analyze user preferences, suggest design elements, and optimize models based on historical data

AI and NLP

- Natural Language Processing (NLP) is a discipline within AI that combines linguistics and computational technology. It focuses on enhancing machines' ability to **understand and generate human language**
- Advancements in language generation models have been driven by techniques like recurrent neural networks (RNN) and transformer models

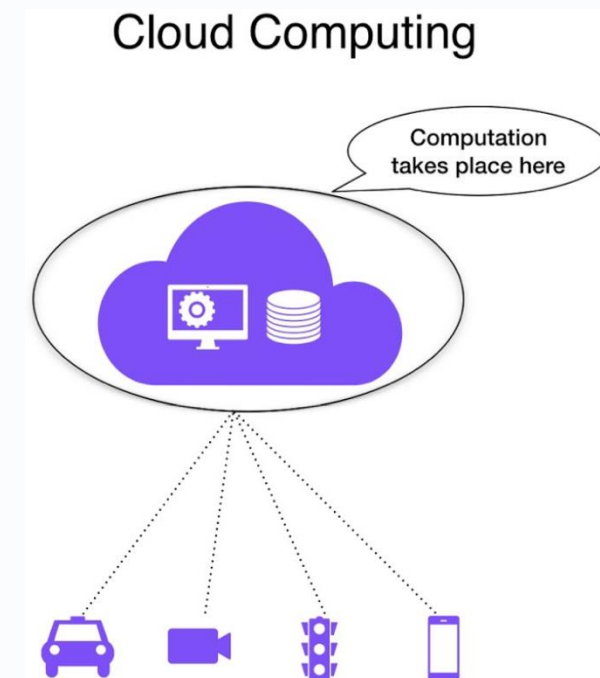
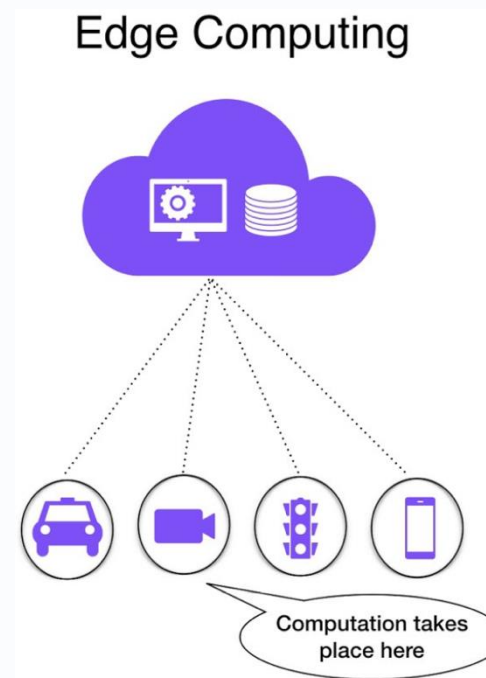
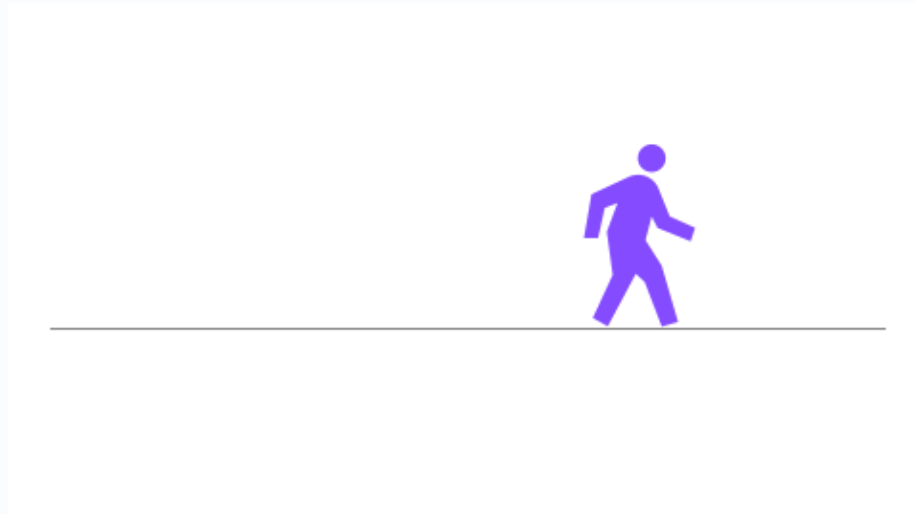
AI and Robotics

- NVIDIA aims to harness expertise in accelerated computing and advanced graphics, thereby empowering robots to sense, think, and act autonomously
- This revolutionary approach outlines remarkable advancements in manufacturing, warehousing, and service robots, which require machines can perceive and navigate the world around them



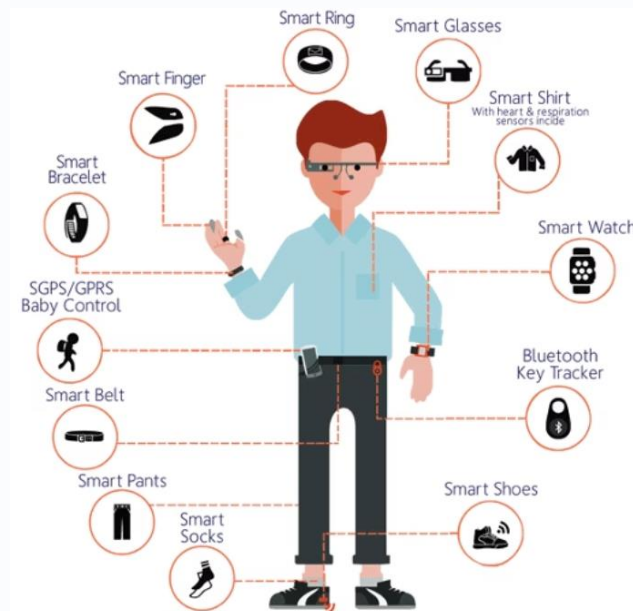
AI and Edge Computing

- **Edge computing** involves processing data near the source of data generation rather than relying on a centralized cloud server. It brings computational power closer to where data is produced, reducing latency and improving efficiency
- It is crucial for AI applications that require real-time processing and low latency



AI and Wearables

- The evolution behind wearable devices is driven by IoT in healthcare, which is known as the Internet of Health Things (IoHT), where medical devices and applications are connected using AI to generate insights from the collected data
- Examples include wearables for Preventive Health, Medication Management, and Tracking Pulse and Blood Pressure for Stress Monitoring



- https://www.bilibili.com/video/BV1HtJmzMERs/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

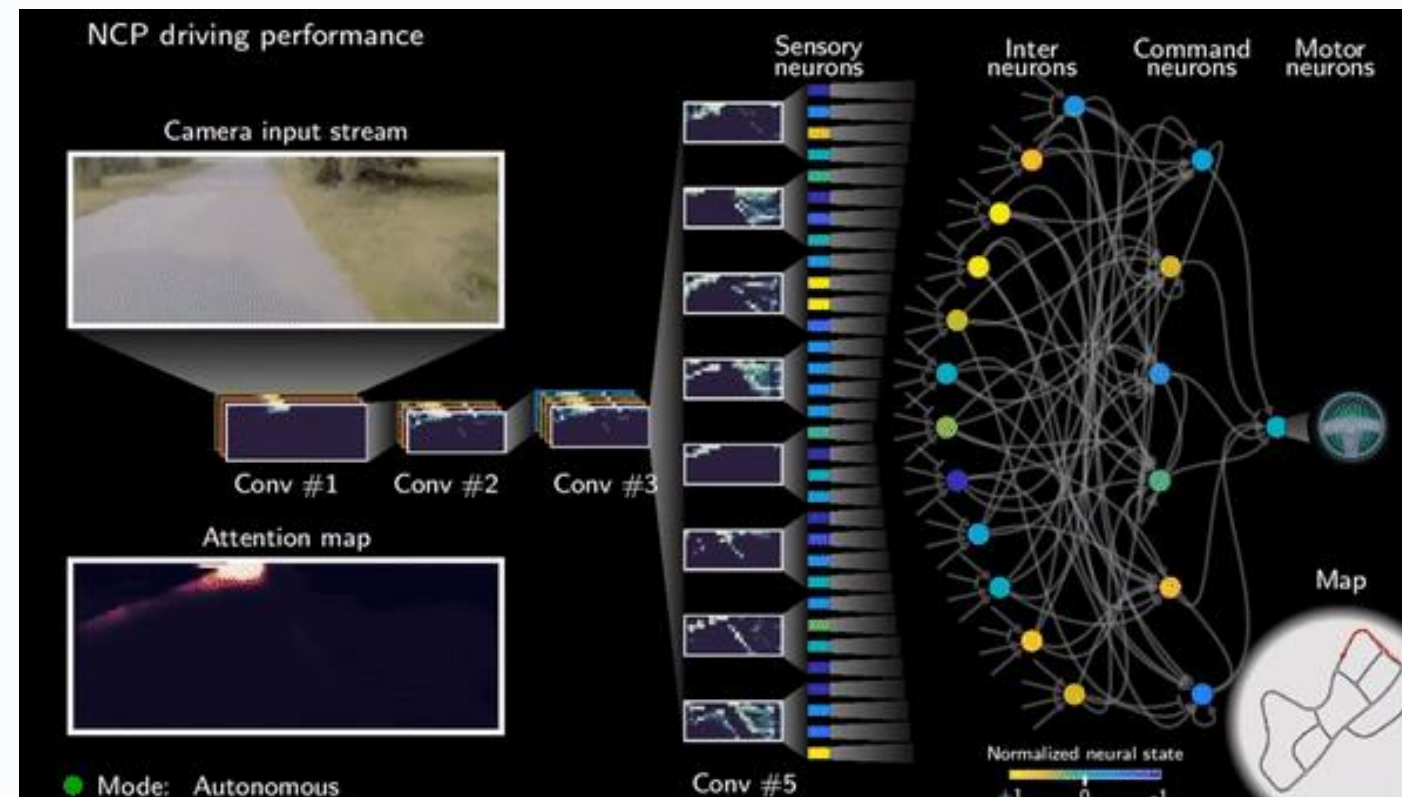
AI and Digital Twins

- A digital twin is a virtual representation of an object or system designed to reflect a physical object.
- It spans the object's lifecycle, is updated from real-time data and uses simulation, machine learning and reasoning to help make decisions
- The fusion of Digital Twins with AI opens up a world of possibilities.



AI and Data-in-Motion

- Data-in-motion refers to the streaming or real-time flow of data from one point to another, which is actively moving and being processed as it travels
- Real-time data is essential for many AI applications, and data-in-motion ensures that the latest information is available for analysis. Streaming data from sensors, devices, or other sources enables AI systems to respond quickly



Fully-developed Techs

Technology	AI Convergence Case	Key Benefits	Challenges
Cloud Computing	Scalable AI training	Cost-efficiency, flexibility	Security risks
GPS/Satellites	Precision navigation	Global coverage, accuracy	Signal interference
Computer Vision	Real-time object detection	Automation in visual tasks	Privacy concerns
NLP	Language understanding	Natural interaction	Language nuances
3D Modeling	Virtual prototyping	Realistic simulation	Computational intensity

Still-evolving Techs

Technology	AI Convergence Case	Key Benefits	Major Challenges
Edge Computing	Real-time AI processing	Low latency, efficiency	Resource constraints
Wearables	Health monitoring	Convenience, real-time data	Battery life, privacy
Robotics	Industrial automation	Precision, speed	Integration complexity
Digital Twins	System simulation	Predictive insights	Data dependency
Data-in-Motion	Streaming analytics	Instant decision-making	Processing complexity

Emerging Techs

Technology	AI Convergence Use Case	Key Benefits	Major Challenges
Quantum Computing	Complex optimization	Exponential speed	Technical complexity
Neuromorphic Computing	Cognitive AI	Low power, efficiency	Early-stage R&D
Advanced Robotics	Human-AI collaboration	Precision, adaptability	High costs

Sources

- <https://medium.com/swlh/neurosymbolic-ai-to-give-us-machines-with-true-common-sense-9c133b78ab13>
- <https://smythos.com/developers/agent-development/symbolic-ai-vs-statistical-ai/>
- <https://levelup.gitconnected.com/the-evolution-of-ai-from-symbolic-to-neural-3fbe384ccaf0>
- <https://www.sciencedirect.com/science/article/pii/S0952197624015276>
- <https://www.linkedin.com/pulse/convergence-ai-geospatial-data-climate-change-research-bhoda-zqezc>
- <https://www.linkedin.com/pulse/convergence-ai-geospatial-data-climate-change-research-bhoda-zqezc>
- <https://medium.com/@3dailyai/ai-3d-modeling-the-next-gen-of-3d-rendering-design-fad254360cd5>
- <https://www.linkedin.com/pulse/convergence-ai-robotics-unveiling-future-nvidia-gtc-erin-rapacki-1ggic>
- <https://www.clouddatainsights.com/the-convergence-will-be-a-powerful-trend-in-technology/>

Questions?

Course Feedback

